

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



INTERNSHIP REPORT ON

“Building an AI-powered Financial Research Co-Pilot”

Submitted in partial fulfillment for the award of the degree of

Bachelor of Engineering

In

Artificial Intelligence and Machine Learning

Submitted by:

Gurudarshan L - 1DS21AI018

Undertaken at



No. 19, 4th C Cross, Koramangala Industrial Area, Koramangala VI Bk, Bangalore South,
Bangalore- 560095, Karnataka

**Department of Artificial Intelligence and Machine Learning,
Dayananda Sagar College of Engineering, Bengaluru**

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India



CERTIFICATE

This is to certify that the Internship titled **“Building an AI-powered Financial Research Co-Pilot”** carried out by **Gurudarshan L [1DS21AI018]**, a bonafide student of **Dayananda Sagar College of Engineering**, in partial fulfillment for the award of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** during the academic year **2024–2025**. It is certified that all corrections/suggestions indicated have been incorporated in the report. The internship report has been approved as it satisfies the academic requirements in respect of the course Internship (21INT82).

Signature of the
Reporting Manager
Sahil Shubam,
Full Stack Engineer,
Finsynth

Signature of the
Internship Coordinator
Prof. Kavya D N,
Assistant Professor,
Dept. of AI & ML, DSCE

Signature of the
Head of the Department
Dr. Vindhya P Malagi,
Professor & Head,
Dept. of AIML, DSCE

Name of the Examiners

1.

2.

Signature with date

.....

.....

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



DECLARATION

I, **Gurudarshan L**, a final year student of **Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bengaluru**, hereby declare that this internship report is based on the work undertaken during my ongoing internship in the role of **Intern, Full Stack Intern** at **Finsynth, Bengaluru**.

This report is submitted in partial fulfillment of the requirements for the award of the Bachelor's Degree in Artificial Intelligence and Machine Learning during the academic year 2024–2025.

Date: 12/04/2025

Place: Bengaluru

Finsynth Technologies Private Limited

CIN: U62099KA2023PTC180644

No. 19, 4th C Cross, Koramangala Industrial Area, Koramangala VI Bk, Bangalore South, Bangalore- 560095, Karnataka

4th March 2025

Dear Gurudarshan L,

We are pleased to offer you an internship at our company as a **Full Stack Intern**. The details of your internship with the Company are set forth below:

1. You are eligible for a total stipend of **₹35,000 per month** during the term which shall be paid on completion of the tasks assigned to you during your internship to the satisfaction of the Company.
2. The internship will be in person and the location will be Accel LaunchPad, 881, 6th Cross Rd, Koramangala 6th Block, Bengaluru, Karnataka 560095.
3. The duration of the internship will be **upto 6 months**.
 - a. During the semester, the internship is **subject to obtaining suitable NoC** from the relevant college authority to continue working in person from our office location.
4. You are expected to work for **atleast 8 hours a day** from our office in Koramangala
5. You will **sign a confidentiality agreement** with the company before you commence your internship.
6. The internship cannot be construed as employment or an offer of employment with FinSynth Technologies Private Limited.

Please confirm your acceptance of the terms of this offer by **4th March 2025** failing which, we have the right to cancel the internship. We look forward to having you on our team! If you have any questions, feel free to contact us.

ACKNOWLEDGEMENT

This internship is the result of continuous guidance, support, and encouragement from several individuals, to whom I am deeply grateful. I take this opportunity to express my sincere appreciation to everyone who contributed to the successful completion of this internship.

I would like to express my heartfelt thanks to **Dr. B G Prasad**, Principal, Dayananda Sagar College of Engineering, for providing the necessary facilities and support to undertake this internship.

My sincere gratitude goes to **Dr. Vindhya P Malagi**, Head of the Department – Artificial Intelligence & Machine Learning, for providing me the opportunity to pursue this internship and for her valuable guidance and encouragement throughout.

I am thankful to **Prof. Kavya D N**, Internship Coordinator and Assistant Professor – AI & ML, for her continuous support, timely suggestions, and motivation during the internship period.

I would also like to extend my thanks to **Asst Prof. Kavya D N**, internal guide, for her consistent guidance, timely feedback, and unwavering support throughout the internship. Her mentorship helped me stay aligned with both academic and industry expectations.

I would like to thank **Sahil Shubam**, Full Stack Engineer, Finsynth for the valuable industry insights and mentorship provided during my role as an intern in Full Stack Intern.

My thanks also go to all the faculty members and non-teaching staff of the Department of AI & ML for their cooperation and assistance during the internship period.

Gurudarshan L [1DS21AI018]

TABLE OF CONTENTS

Sl no	Description	Page No
1	Introduction	1
2	About the Company	2
3	Scope of the Work	3
4	Learning Objectives	5
5	Challenges & Solutions	7
6	Achievements & Contributions	8
7	Reflection and Analysis	10
8	Recommendations	11
	Conclusions	12

CHAPTER 1

INTRODUCTION

I had the opportunity to work as a Full Stack Intern at **Finsynth**, an AI-driven fintech startup focused on transforming the way financial analysts conduct research. Finsynth is building an **AI-powered financial research co-pilot**, specifically designed to assist analysts who employ **long-short trading strategies**. The product leverages artificial intelligence to streamline and accelerate the analysis of companies by integrating structured and unstructured financial data into a seamless research experience.

Under the mentorship of **Sahil Shubam**, a Full Stack Engineer at Finsynth, I contributed across multiple layers of the tech stack. My internship primarily focused on **web development**, **Excel plugin integration**, and **backend (gateway) enhancements**. I worked closely with real-world codebases, participated in team discussions, and tackled bugs and feature requests with production-level responsibility.

Throughout the internship, I used a modern technology stack that included **TypeScript**, **React**, **Fluent UI**, **Apollo GraphQL**, and **TanStack Router** on the frontend. On the backend, I worked with **TypeScript**, **GraphQL**, and **Codegen** to build and maintain the gateway. My work included implementing user-facing features like the company profile page and ribbon buttons in the Excel plugin, integrating APIs, and translating Figma designs into responsive UI components.

In addition to development, I gained hands-on experience with **bundlers**, **GitHub Actions**, **workflows**, **IndexedDB**, and **request workers**. These tools and practices gave me a deeper understanding of full-stack development in a fast-paced, agile environment.

This internship has significantly strengthened both my technical skills and understanding of how AI can empower financial workflows, and has provided a solid foundation for my future in software development.

CHAPTER 2

ABOUT THE COMPANY



Finsynth is a cutting-edge fintech startup on a mission to revolutionize the way financial research is conducted. At its core, Finsynth is building an **AI-powered financial research co-pilot** that enables equity analysts, especially those leveraging **long-short trading strategies**, to analyze companies quickly, intelligently, and with greater context. Traditional financial analysis is time-consuming and fragmented—analysts must navigate through earnings reports, SEC filings, market data, and expert commentary from multiple sources. Finsynth aims to centralize and streamline this process using artificial intelligence.

The platform uses advanced natural language processing (NLP), knowledge graphs, and real-time data integration to provide a unified research environment. It combines structured financial metrics with unstructured data such as transcripts, filings, and news to offer insights that are not only comprehensive but contextually rich. By doing this, Finsynth dramatically reduces the manual effort required in analyzing a company's fundamentals and accelerates the decision-making process for investors.

Finsynth is designed to be highly interactive and user-centric, incorporating modern UI/UX design and seamless integration with familiar tools like Microsoft Excel. This allows analysts to stay within their preferred workflows while benefiting from powerful AI augmentation.

The team at Finsynth is made up of talented engineers, designers, and finance professionals who bring a deep understanding of both technology and capital markets. The company embraces an agile development culture, where iterative feedback, rapid prototyping, and cross-functional collaboration are key to product evolution.

As a full stack intern, I had the opportunity to work closely with the product, gaining firsthand exposure to how such a transformative tool is being built. Finsynth's vision to empower financial analysts through intelligent software is not just innovative—it's actively shaping the future of financial research.

CHAPTER 3

SCOPE OF THE WORK

During my internship at Finsynth, I had the opportunity to work across multiple domains within the product ecosystem. My responsibilities spanned frontend and backend development, Excel plugin integration, and tooling improvements. Each area contributed to a deeper understanding of full-stack development in a real-world, fast-paced startup environment. Below are the key segments of my work:

1. Excel Plugin Development

One of my initial responsibilities was contributing to the **Excel plugin**, a core part of Finsynth's product experience. The plugin acts as a bridge between the Finsynth platform and analysts' preferred research tool—Microsoft Excel. I worked on implementing **custom ribbon buttons**, enabling users to perform contextual actions, fetch company data, and trigger model updates directly from Excel. I also helped fix bugs related to data rendering and ensured smooth API communication between the plugin and backend services.

2. Web Application Frontend

The web application serves as the main interface for users to interact with company data, research insights, and AI-powered analysis. On the frontend, I used **TypeScript, React, Fluent UI, TanStack Router**, and **Apollo GraphQL** to build and refine features. I developed user interfaces such as the **company profile** section, contributed to page routing, and translated complex **Figma designs** into responsive and pixel-perfect UI components. This part of the work also included handling client-side caching and optimizing the user experience.

3. Gateway & Backend Development

On the backend, I worked with the **GraphQL gateway**, built using **TypeScript** and **Codegen**. My role involved integrating third-party APIs, modifying GraphQL schemas, and connecting frontend queries to backend data sources. I also fixed bugs, added new resolvers, and contributed to ensuring the type-safety and consistency of the data flow. Codegen played a crucial role in maintaining strong typing across services and reducing manual effort when updating schemas or query structures.

4. Dev Tools & Infrastructure

Beyond core development, I explored and contributed to the underlying infrastructure that supports smooth engineering workflows. I learned to configure and troubleshoot **GitHub Actions** for CI/CD pipelines, used **bundlers** like Webpack and Vite for optimizing builds, and worked with **IndexedDB** for client-side storage in the browser. I also gained hands-on experience with **request workers**, which are used to manage data fetching logic more efficiently and reduce API response times.

5. Collaboration & Agile Practices

Throughout the internship, I was part of daily stand-ups, sprint reviews, and async discussions using tools like Slack, Linear, and GitHub. I collaborated closely with my mentor and other engineers to break down features, review code, and ensure that tasks aligned with user requirements and design expectations. This helped me understand how cross-functional teams work together to deliver high-quality software in an agile development environment.

CHAPTER 4

LEARNING OBJECTIVES

Throughout my internship at Finsynth, I was able to achieve significant progress in both technical and professional skills. Below are the key learning objectives I aimed to accomplish, and how they were met during my internship.

1. Mastering Full-Stack Development

A central learning objective was to gain proficiency in full-stack development, bridging the gap between frontend and backend technologies. Finsynth provided the perfect environment for this. I worked extensively with **TypeScript**, **React**, and **Apollo GraphQL** on the frontend, and **GraphQL** combined with **TypeScript** and **Codegen** on the backend. I developed the technical ability to connect data from APIs to user interfaces, ensuring seamless interactions between the client and server.

The exposure to both frontend and backend processes allowed me to understand the flow of data from backend services all the way to the user interface. I learned how to design APIs using GraphQL, optimize query execution, and handle responses on the client side in React. Additionally, my work on **Excel plugin integration** provided insights into extending web applications into desktop environments, while ensuring performance and functionality remain top-notch.

2. Gaining Expertise in UI/UX Design and Implementation

Another key objective was to refine my understanding of **UI/UX design principles** and their practical implementation. I worked closely with Finsynth's designers to translate **Figma designs** into working React components using **Fluent UI**. This involved maintaining consistency with design systems, ensuring responsiveness, and applying best practices for accessibility.

By working on high-fidelity UI components like the **company profile page**, **data tables**, and custom **ribbon buttons** in the Excel plugin, I developed a keen eye for design details and usability. I also focused on enhancing user interactions by ensuring smooth page transitions and optimizing component performance, all while adhering to established design systems and style guides.

This process helped me grasp how design and code intertwine to create effective and intuitive user experiences, making me more comfortable with front-end development tools and practices.

3. Understanding and Implementing Agile Development Practices

A significant learning objective was gaining exposure to **Agile development** workflows, particularly in an engineering team. I participated in daily stand-ups, sprint planning, code reviews, and sprint retrospectives. This involvement provided an understanding of how agile methodologies improve team productivity, encourage fast iteration, and help manage priorities effectively.

I learned to work in two-week sprint cycles, ensuring that my tasks were manageable and aligned with the overall goals of the team. By collaborating on features with both developers and designers, I understood the importance of breaking down tasks into smaller, more achievable objectives and ensuring that each step was consistent with the overall product vision.

The experience taught me how to prioritize tasks, manage time efficiently, and adapt quickly to changing requirements, all of which are crucial skills for future development projects in fast-paced environments.

4. Building and Maintaining CI/CD Pipelines

During my internship, I sought to deepen my understanding of **Continuous Integration/Continuous Deployment (CI/CD)** practices. By contributing to **GitHub Actions** workflows, I gained hands-on experience automating testing, linting, and deployment processes. I learned how to configure workflows for different environments, ensuring that code is properly tested and deployed without manual intervention.

I also worked with **bundlers** such as Webpack and Vite to streamline build processes, enabling faster feedback loops during development. Learning how to troubleshoot build errors, optimize assets, and configure the project for multiple deployment environments (development, staging, production) enhanced my understanding of the software delivery lifecycle.

This objective helped me recognize the importance of automation in modern software development, improving code quality and minimizing errors during deployments.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Throughout my internship at Finsynth, I encountered a range of technical and non-technical challenges that tested my adaptability, problem-solving abilities, and communication skills. Each of these hurdles presented a learning opportunity, and overcoming them helped me grow into a more confident and capable full-stack developer. Below are some key challenges I faced and the solutions that helped me move forward.

1. Understanding a Complex Codebase

When I joined Finsynth, one of the initial hurdles was navigating the existing codebase. As the product already had several interconnected components—such as the Excel plugin, web application, and gateway backend—it was initially overwhelming to grasp how everything worked together. The depth of TypeScript usage, combined with GraphQL schemas and auto-generated code, meant I had to spend considerable time reading, tracing logic, and understanding project structure. However, by focusing on smaller, manageable tasks and learning progressively through code reviews and pair programming sessions, I gradually built a solid mental model of the system. Collaborating with other developers and following consistent patterns across the stack helped ease this learning curve significantly.

2. Debugging Cross-Layer Issues

In a full-stack system, bugs often don't originate in the layer where they appear. I encountered several situations where a UI issue was actually caused by incorrect data handling in the backend or mismatched query parameters in GraphQL. These bugs required a full-stack debugging mindset, and I had to systematically inspect each part—from the frontend React components to the GraphQL queries, and all the way down to the backend resolvers and external APIs. Over time, I developed a methodical approach: using browser DevTools, GraphQL playground, and detailed logs to trace the flow of data. This experience sharpened my debugging skills and made me more comfortable handling issues that span multiple layers.

3. Integrating APIs and Maintaining Type Safety

Integrating new APIs into the gateway was another key challenge. Each external API had different response structures, and mapping them accurately to our GraphQL schema required careful transformation and validation. Ensuring consistency across the backend and frontend was especially important, given the type-safe environment powered by Codegen. It took time

to understand how resolvers were structured, how fragments were used, and how errors should be handled gracefully. Once I became familiar with the conventions, I was able to confidently integrate third-party APIs, adapt them to our needs, and ensure that type mismatches or unexpected behaviors were minimized.

4. Translating Figma Designs into Interactive Components

Turning detailed Figma designs into working, responsive UI components was a recurring responsibility. While Fluent UI offered a set of powerful components, it also came with its own learning curve, especially when it came to custom styling and layout behavior. Some components needed to match pixel-perfect designs while remaining adaptable across screen sizes. I had to understand spacing, typography, hover states, and UI patterns in depth to ensure consistency with design expectations. Through consistent iteration, attention to detail, and feedback from designers, I improved my ability to translate design systems into clean, maintainable code.

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

1. Feature Development Across Platforms

During my internship, I had the opportunity to work across Finsynth's Excel plugin, web application, and backend gateway. I contributed to several core features such as the company profile section, ribbon buttons in the Excel plugin, and the model update functionality. Each feature I worked on helped enhance the product's overall usability and allowed analysts to interact more seamlessly with the AI-powered co-pilot.

2. End-to-End API Integration

One of my key contributions was integrating multiple APIs from backend to frontend using GraphQL. I worked on defining GraphQL queries, integrating them into the gateway, and consuming them effectively in the frontend with Apollo Client. Ensuring smooth data flow and type safety across the stack helped deliver reliable features to users.

3. Translating Figma Designs into Functional UI

I took ownership of implementing UI components based on Figma designs. By using React, Fluent UI, and Tailwind principles where necessary, I translated static mockups into fully responsive and interactive elements. My attention to design detail ensured that the UI remained consistent with the product's aesthetic and user expectations.

4. Code Quality and Debugging

Throughout the internship, I handled multiple bug fixes across different components, gaining a strong understanding of the codebase. Whether it was a UI glitch or a deeper issue involving IndexedDB or request workers, I contributed by debugging, testing, and pushing clean, functional code that improved app stability.

5. Learning and Adapting to Workflow Tools

I quickly ramped up on key workflow tools like GitHub Actions, bundlers, and automated workflows. My adaptability and ability to independently understand these tools allowed me to contribute efficiently to CI/CD processes and maintain high code quality in production deployments.

CHAPTER 7

REFLECTION AND ANALYSIS

Interning at Finsynth has been a transformative experience that allowed me to bridge the gap between academic knowledge and practical application. From the moment I joined, I was immersed in a fast-paced, product-oriented environment where real-world challenges required immediate and thoughtful solutions. Working on a cutting-edge AI-powered financial research co-pilot exposed me to a unique intersection of finance, technology, and user experience. This internship not only strengthened my technical skills but also shaped my mindset as a developer working within a collaborative and outcome-driven team.

One of the most valuable aspects of this experience was the opportunity to work full stack across multiple platforms—ranging from the Excel plugin to the web application and backend gateway. This gave me a holistic view of how different components of a product communicate and function together. I gained a better understanding of how a request travels from the user interface to the backend, how data is queried and transformed, and how design and performance trade-offs are made at each level.

Collaboration played a significant role in my growth. Working under the guidance of Sahil Shubam and being a part of a responsive and knowledgeable team pushed me to improve my communication, problem-solving, and code review practices. Whether it was receiving feedback on a pull request or pair programming to debug a tricky issue, every interaction was a learning opportunity.

Another critical realization was how important it is to write maintainable, scalable, and readable code in a collaborative environment. Unlike academic projects, where quick solutions often suffice.

Overall, the internship provided not only technical depth but also a broader understanding of how software is developed, tested, and deployed in a real-world setting. It gave me clarity about the kind of work I enjoy—building thoughtful user experiences backed by solid, well-structured systems—and affirmed my interest in continuing to grow as a full stack developer in the tech industry.

CHAPTER 8

RECOMMENDATIONS

Based on my internship experience at Finsynth, I would like to offer several recommendations that may help improve the internship process for future interns and further enhance team productivity and learning outcomes. These suggestions stem from my observations, day-to-day work, and overall reflections during the internship.

Firstly, having a brief onboarding guide or technical documentation tailored specifically for interns would be highly beneficial. While I eventually became comfortable with the codebase and workflows, the initial days involved a lot of exploration and guesswork. A structured onboarding document that covers common tools (e.g., GraphQL, Apollo, Codegen), workspace setup, and coding conventions could significantly reduce the ramp-up time and improve early-stage productivity.

Secondly, scheduling a weekly or bi-weekly check-in with mentors or senior developers can offer structured guidance and create space for sharing blockers, clarifying doubts, and receiving feedback. While I did receive support from my mentor whenever needed, having consistent checkpoints would make the learning curve smoother and encourage proactive knowledge sharing.

Another recommendation would be to involve interns in small-scale planning or feature refinement meetings. This exposure helps develop product thinking and gives interns context on how and why certain technical decisions are made. Understanding the ‘why’ behind the work significantly improves how someone approaches the ‘how’.

In terms of learning, offering curated internal resources or a mini roadmap tailored to the tech stack can empower interns to self-learn and contribute more confidently. For example, a reading list on Fluent UI patterns, GraphQL query design, or GitHub Actions setup would provide direction during downtime or onboarding phases.

Overall, these recommendations are aimed at creating a more structured, encouraging, and growth-focused environment that benefits both the intern and the organization.

CHAPTER 9

CONCLUSION

My internship at Finsynth has been an enriching journey, both professionally and personally. It provided me with a real-world platform to apply and extend my technical skills while also helping me understand the intricacies of working in a fast-paced, startup environment. Being part of a team building an AI-powered financial research co-pilot was not only intellectually stimulating but also a unique opportunity to contribute to a product at the intersection of finance, technology, and user experience. I was able to work hands-on across the stack—contributing to the Excel plugin, web application, and backend gateway—allowing me to grow into a more holistic developer.

Throughout this journey, I got the chance to develop and ship several key features, fix critical bugs, and actively participate in code reviews and collaboration. I learned how to translate Figma designs into functional UI components, build reliable APIs using GraphQL, manage frontend state with Apollo Client, and understand the inner workings of tools like GitHub Actions and bundlers. More importantly,

Working under the mentorship of Sahil Shubam and alongside a driven team helped me gain confidence, discipline, and a deep appreciation for clean, scalable code. I also developed soft skills like communication, ownership, and proactive problem-solving, which are crucial in any professional setting. Every challenge I faced was met with support and trust, creating a space where I could experiment, fail, learn, and grow.

This internship has been a milestone in my career, reaffirming my passion for building impactful digital products and strengthening my desire to pursue a future in full stack development. I leave Finsynth with not just new technical skills but also with a better understanding of what it means to contribute meaningfully to a product and team. I am grateful for this experience and confident that the knowledge and mindset I've gained will guide me well in the next phase of my journey.

.